

INCORRECT

Yes

(B) No

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Got it, thanks! ^

Suppose $a_n = f(n)$ and f is positive, continuous, and decreasing for $x \geq N$, the integral test says that

$$\sum_{n=N}^{\infty} a_n \text{ and } \int_N^{\infty} f(x) dx$$

either both converge or both diverge.

$$\frac{\sqrt{x}}{\ln(x)}$$

is continuous and positive for all $x \geq 2$.

To find whether it's always decreasing for $x \geq 2$, let's consider its derivative.

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$$\frac{d}{dx} \left(\frac{\sqrt{x}}{\ln(x)} \right) = \frac{\ln(x) - 2}{2\sqrt{x} \ln^2(x)}$$

Got it, thanks! ^

$$\begin{aligned} & \frac{d}{dx} \left(\frac{\sqrt{x}}{\ln(x)} \right) \\ &= \frac{\ln(x) \frac{d}{dx} [\sqrt{x}] - \frac{d}{dx} [\ln(x)] \sqrt{x}}{\ln^2(x)} \quad \text{Quotient rule} \\ &= \frac{\ln(x) \left(\frac{1}{2\sqrt{x}} \right) - \frac{1}{x} \cdot \sqrt{x}}{\ln^2(x)} \\ &= \frac{\sqrt{x} \ln(x) - 2\sqrt{x}}{2x \ln^2(x)} \\ &= \frac{\ln(x) - 2}{2\sqrt{x} \ln^2(x)} \end{aligned}$$

For $x > e^2$, we have $\ln(x) > 2$, which means $\ln(x) - 2$ is positive.

So $\frac{d}{dx} \left(\frac{\sqrt{x}}{\ln(x)} \right)$ is positive for all $x > e^2$, which means $\frac{\sqrt{x}}{\ln(x)}$ is

